SEC:: SR. JEE MAINS APEX | GTM-23 | Date: 22.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	B	C	B	B	A	B	D	C	C
11	12	13	14	15	16	17	18	19	20
A	B	B	A	B	D	C	A	C	C
21	22	23	24	25					
100	25	220	9	10					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	D	B	C	B	B	A	C	A	D
36	37	38	39	40	41	42	43	44	45
A	C	A	B	B	C	B	A	C	B
46	47	48	49	50					
2	8	6	4	10					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	C	B	C	B	B	A	B	C	D
61	62	63	64	65	66	67	68	69	70
D	C	B	D	D	C	C	B	D	C
71	72	73	74	75					
2	27	17	3	3					

01. Critical angle between glass and liquid face is

$$\sin\theta_c = \frac{3/2}{\mu} = \frac{3}{2\mu} \qquad \ldots(i)$$

Angle of incidence at face AC is $60°$

i.e., $i = 60°$

For total internal reflection to take place,

$i > \theta_c$

or $\sin i > \sin\theta_c$

or $\sin 60° > \frac{\mu}{3/2}$

02. According to question,

$\sigma_1 = \sigma_2 = \sigma_3$

$\because \quad \sigma = \frac{Q}{4\pi r^2}$

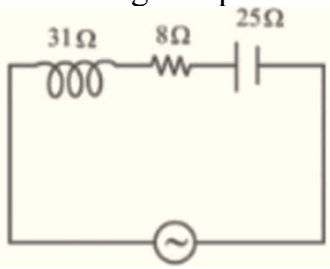
$\because \quad = \frac{Q_1}{4\pi r^2} = \frac{Q_1 + Q_2}{4\pi\left(4R^2\right)} = \frac{Q_1 + Q_2 + Q_3}{4\pi\left(9R^2\right)}$

$Q_1 = \frac{Q_1 + Q_2}{4} = \frac{Q_1 + Q_2 + Q_3}{9}$

On solving the above equation, we get - $Q_1 : Q_2 : Q_3 = 1:3:5$

03. Given that, $X_L = 31\Omega, R = 8\Omega \&, X_C = 25\Omega$

According to equation



We know that, impedance of circuit –

$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2}$

$Z = \sqrt{8^2 + (31 - 25)^2}$

$Z = \sqrt{64 + 36}$

$Z = 10$

$\therefore$ Power factor,

$\cos\phi = \frac{R}{Z} = \frac{8}{10}$

$\text{Cos}\phi = 0.8$

04. Given,

$m_2 = \frac{m_1}{2}, (K.E)_2 = 16(K.E)_1$

We know,

Kinetic energy (K.E) = $\frac{p^2}{2m}$

Therefore, The ratio of $\frac{(K.E.)_2}{(K.E.)_1} = \frac{p_2^2 / 2m_2}{p_1^2 / 2m_1} = \frac{2p_2^2}{p_1^2}$

$\frac{16(K.E.)_1}{(K.E.)_1} = \frac{2p_2^2}{p_1^2}$

$p_2^2 = 8p_1^2$

$p_2 = 2\sqrt{2}p_1$

05. $E = W_0 + K_{\max}$ .... (i)

$\Rightarrow hf = W_A + K_A$ .....(ii)

and $2hf = W_B + K_B = 2W_A + K_B \left(\because \frac{W_A}{W_B} = \frac{1}{2} \right)$

Dividing equation (i) by (ii)

$\frac{1}{2} = \frac{W_A + K_A}{2W_A + K_B} \Rightarrow \frac{K_A}{K_B} = \frac{1}{2}$

06. $e = \frac{\phi_i - \phi_f}{\Delta t}$

$\Rightarrow \Delta t = \frac{BA - 0}{e} = \frac{20 \times 10^{-2}}{10} = 20ms$

07. $P_1 = 3f_c = 3\left(\frac{v}{4l_c} \right)$

$P_2 = 4f_o = 4\left(\frac{v}{2l_o} \right)$

Where l_c and l_o are lengths of closed and open organ pipe respectively.

Since both are in resonance, so

$3\left(\frac{v}{4l_c} \right) = 4\left(\frac{v}{2l_o} \right)$

$\Rightarrow \frac{l_c}{l_o} = \frac{3}{8}$

08. Length of each wire : l; Area of thick wire (A_1) = 3A; Area of thin wire (A_2) = A and resistance of thick wire (R_1) = $10\,\Omega$

Resistance (R) = $\rho\frac{l}{A}$

$$\frac{R_1}{R_2}=\frac{A_2}{A_1}=\frac{A}{3A}=\frac{1}{3}$$

$$\Rightarrow R_2=3R_1=3\times10=30\Omega$$

The equivalent resistance of these two resistors

In series = $R_1+R_2=30+10=40\Omega$

09. The correct option is C 1, 0, 1

The output of OR gate is

Y' = (A + B)

The output and AND gate is

Y = Y' . C = (A + B) . C

If A = 1, B = 0, C = 1, they Y = (1 + 0). 1 = 1

10. $\overline{E}=\overline{V}\times\overline{B}$

$$\overline{E}=3\times10^{8}\times2\times10^{-8}$$

$$\overline{E}=6\hat{j}$$

As $\overline{E}\ \&\ \overline{B}$ remains perpendicular

11. Transition from $4E$ to E

$$\left(4E-E\right)=\frac{hc}{\lambda_1}\Rightarrow\lambda_1=\frac{hc}{3E} \qquad \ldots.(i)$$

Transition from $\frac{7}{3}E$ to E

$$\left(\frac{7}{3}E-E\right)=\frac{hc}{\lambda_2}\Rightarrow\lambda_2=\frac{2hc}{4E} \qquad \ldots.(ii)$$

From equation (i) and (ii): $\frac{\lambda_1}{\lambda_2}=\frac{4}{9}$

12. $v_e=\sqrt{\frac{2GM}{R}}=R\sqrt{\frac{8}{3}\pi G\rho}$

If mean density if constant, then $v_e\propto R$

$$\frac{v_e}{v_p}=\frac{R_e}{R_p}=\frac{1}{2}\Rightarrow v_e=\frac{v_p}{2}$$

13. Magnetic susceptibility, $x = \mu_r - 1$

Or $x = \dfrac{\mu}{\mu_0} - 1$

Or $499 = \dfrac{\mu}{4\pi \times 10^{-7}} - 1$

Or $\mu = 500 \times 4\pi \times 10^{-7}$

$= 2\pi \times 10^{-4}\ Hm^{-1}$

14. $W_{net} = -V\ e \text{ and So } Q_{net} = -V\ e$

$\Delta U = 0$ for full cycle.

15. $\Rightarrow Current\ in\ 6\Omega = \dfrac{8}{6} = \dfrac{4}{3}\ A = 1.33\ A$

16. $\lambda = \dfrac{h}{\sqrt{2mVq}}$

17. $\dfrac{B}{E} = \dfrac{1}{C}\ \&\ \mu_0 \varepsilon_0 = \dfrac{1}{c^2}$

$\therefore \dfrac{B}{E}.\mu_0 \varepsilon_0 = \dfrac{1}{c^3} = M^0 L^{-3} T^3$

18. $P_{Excess} = h\rho g + \dfrac{2s}{r}$

19. Time taken by water to reach the bottom

$= t = \sqrt{\dfrac{2(H-D)}{g}}$

and velocity of water coming out of hole, $v = \sqrt{2gD}$

$\therefore$ Horizontal distance covered $x = v \times t$

$= \sqrt{2gD} \times \sqrt{\dfrac{2(H-D)}{g}} = 2\sqrt{D(H-D)}$

20. $P_i = mv_0 \hat{j}$

After shoots momentums of system will remain conserved.

$\vec{P}_f = \dfrac{2}{3} m\vec{v} + \dfrac{m}{3}(2v_0)\hat{i}$

$\vec{P}_i = \vec{P}_f$

$mv_0\hat{j} = \frac{2}{3}m\vec{v} + \frac{2}{3}mv_0\hat{i}$

$\vec{v} = \frac{3}{2}v_0\hat{j} - v_0\hat{i}$

$|\vec{v}| = \sqrt{\frac{9}{4}v_0^2 + v_0^2} = \frac{\sqrt{13}}{2}v_0$

21. For 9th turn

$2a = 2 \times 50 = 100$

22. Resultant intensity $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\theta$

If path difference, = λ then phase difference = 2π ..... $\left[\because \phi = \frac{2\pi}{\lambda} \times \Delta x\right]$

$\therefore I_R = I + I + 2\sqrt{1 \times 1}\cos 2\pi$ .....$\left(\because I_1 = I_2 - I\right)$

$= 4I = M$

If path difference = $\frac{\lambda}{3}$

Then phase difference $= \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3} rad$

$I_R^{'} = I + I + 2\sqrt{I \times I}\cos\frac{2\pi}{3}$

$= 2I + 2I\left(\frac{-1}{2}\right) = I = \frac{M}{4} = 0.25M$

23. $^{200}X \rightarrow {}^{80}Y + {}^{120}Z$

6.5 7 8

Energy released = 80 x 7 + 120 x 8 – 200 x 6.5

= 220 MeV

= 2200 x 10^5 eV

24. According to Law of conservation of energy

$\therefore mgh = \frac{1}{2}mv^2$

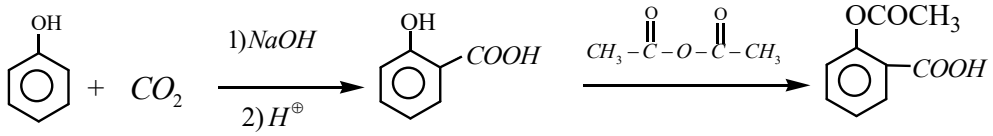
$h = \frac{v^2}{2g}$

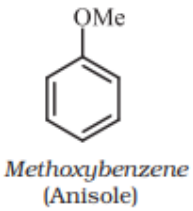
$\frac{X}{5} = \frac{36}{2 \times 10} = 1.8$

(rotational K.E. remain same)

25. $y_{cm} = \frac{ma}{10m} = \frac{a}{10}$

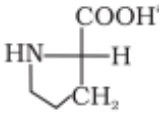
CHEMISTRY

26. Statement I and Statement II are true. Noble gases show large positive electron gain enthalpy.
27. Intensity of red colour decreases due to backward reaction by the formation of $[Hg(SCN)_4]^{2-}$
28. acetone and chloroform are mixed together, H-bonding takes place between them. Such a liquid pair show negative deviation from Raoult's law
29. $FeSO_4.7H_2O\ crystals---Pale\ green$, also called green vitriol.
30. All the statements are correct as per NCERT text.
31. Cylopentadienyl anion is an aromatic compound**.**
32. $\Lambda_{eq} = \frac{1000\kappa}{N} = \frac{1000\times 26\times 10^{-2}}{2} = \frac{260}{2} = 1.3\times 10^2\ S\ cm^2 eq^{-1}$
33. 

34. NCERT Statements.
35. In isoelectronic species as positive charge increases size decrease and as negative charge increases size increases.
36. Chromyl chloride is formed only with dry salt and not with aqueous solutions.
37. Spots from dye A matches in distance with blue(B) and green(G)
38. $r = K[NOBr_2][NO] ----1$
    $but\ [NOBr_2] = Kc[NO][Br_2]$
    Therefore, $r = K[NO]^2[Br_2]$
39. pK_b value of Benzenamine is (9.38) aromatic amine while that of Phenylmethanamine is (4.70) aliphatic amine. Hence Benzenamine is weak base than Phenylmethanamine.
40. 

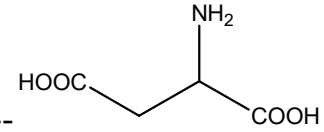
Methoxybenzene
(Anisole)
41. $[Ni(CO)_4]$ – Tetrahedral, **Diamagnetic**
42. $C_2 - molecule$

43. Achiral amino acid— H_2N-CH_2-COOH

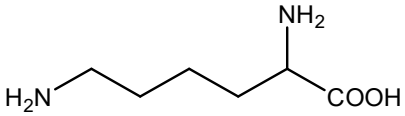


Amino acid with secondary amine group---

Acidic amino acid----



Basic amino acid-----



44. The ability to form $p\pi-p\pi$ multiple bonds among themselves **decreases** down the group

45. Nitrogen—Prussian blue

Sulfur--- $\underset{\text{Violet}}{[Fe(CN)_5NOS]^{4-}}$

Phosphorus---- Yellow Ammonium phosphomolybdate

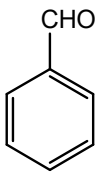
Chlorine--- white ppt- $AgCl$

46. Brown ring formula- $\left[Fe\left(H_2O\right)_5 NO\right]SO_4$

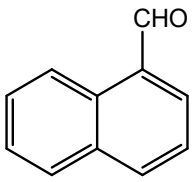
47.

$$\Delta_t=\frac{4}{9}\times\Delta_o=\frac{4}{9}\times 18000=8000=8\times10^3\,cm^{-1}$$

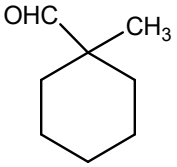
48. $HCHO,\ OHC-CHO$,



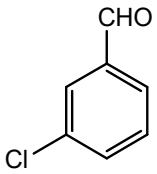
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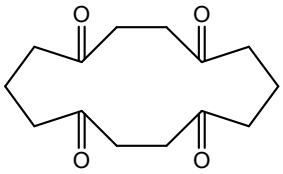
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49. Product is



50. $\Delta G^0=\Delta H^0-T\Delta S^0$

$=-54.07\times1000-298\times10=$ -57.050J

$\Delta G^0=-2.303RT\log K_c$

$-57050=-2.303\times8.314\times298\times\log K_C$

$$\Rightarrow\log K_C=\frac{-57050}{-2.303\times8.314\times298}=\frac{-57050}{-5705}=10$$

$K_C=10^{10}$

MATHEMATICS

51. Assume z = x+iy

x+1=y+8

x=y+7

And $y+8=\sqrt{y^2+(y+7)^2}$

$\Rightarrow y=5 \ \ or \ -3$

52. $\overline{f(x)}=(x-a)(x-c)+\lambda(x-b)(x-d)=0$ then

$f(a)=\lambda(a-b)(a-d)>0$

$f(c)=\lambda(c-d)(c-d)>0$

If $\lambda>0$ then $f(a)>0, f(c)<0 \Rightarrow$ there is a root between a and c.

$S_2: D=0 \quad b^2=4ac$

$\not{4}K^2=\not{4}(K-3)(3K-6)$

$K^2=3K^2-15K+18 \qquad 2K^2-15K+18=0$

$2K^2-12K-3K+18=0$

$2K(K-6)-3(K-6)=0 \quad K=6,\frac{3}{2}$

For K = 3/2 Range $\in(-\infty,0]$

53.

$$I=\int\frac{(e^x+\cos x+1)-(e^x+\sin x+x)}{e^x+\sin x+x}dx=\ln(e^x+\sin x+x)-x+C$$

$$\therefore \qquad f(x)=e^x+\sin x+x$$

and $\qquad g(x)=-x$

$$f(x)+g(x)=e^x+\sin x$$

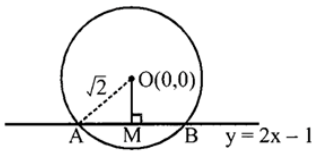
54.

$OA=\sqrt{2}$

$OM=\frac{1}{\sqrt{5}}$

$AM=\sqrt{OA^2-OM^2}=\sqrt{2-\frac{1}{5}}=\frac{3}{\sqrt{5}}$

$AB=2AM=\frac{6}{\sqrt{5}}$



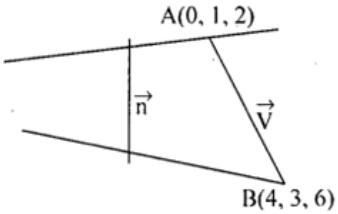
55.

Vector perpendicular to both $\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ 1 & 0 & 2 \end{vmatrix}$

$= \hat{i}(-2) - \hat{j}(6+1) + \hat{k}(0+1)$

$= -2\hat{i} - 7\hat{j} + \hat{k}$ says $2\hat{i} + 7\hat{j} - \hat{k}$

Now, $\vec{V} = \vec{AB} = 4\hat{i} + 2\hat{j} + 4\hat{k}$



Now, $\text{S.D.} = \left| \dfrac{\vec{V} \cdot \vec{n}}{|\vec{n}|} \right| = \left| \dfrac{8+14-4}{\sqrt{54}} \right|$

$= \dfrac{18}{\sqrt{54}} = \dfrac{18}{3\sqrt{6}} = \dfrac{6}{\sqrt{6}}$

$\therefore$ $\text{S.D.} = \sqrt{6} \Rightarrow d = 6$

56. $8C_1 \cdot {}^7C_1 \cdot \dfrac{8!}{2!} = 28(8!)$

57. And the set of all values of p is the interval [a,b)
Now by observation (p, p+1) lies on $x^2+y^2 = 9$
So, solving y-x = 1 and $x^2 + y^2 = 9$, we get

$x^2 + x - 4 = 0$

$x = \dfrac{-1 \pm \sqrt{1+4.4}}{2} = \dfrac{\pm\sqrt{17}-1}{2}$

Hence, $P \in \left[1, \dfrac{\sqrt{17}-1}{2}\right)$

Now on comparing with $p \in (a,b)$ we get

$\Rightarrow a = 1, b^2 + b = 4$

58. Term independent of x

$= {}^{11}C_7 \dfrac{5^{-3}}{2^4} = \dfrac{33}{200}$

59.

$f(x) = e^{g(x)}$

$f'(x) = e^{g(x)} \cdot g'(x)$

$\therefore$ Check the nature of $g(x)$,

$g(x)$ is increasing ($\uparrow$) in $(1, 2) \cup (3, \infty)$

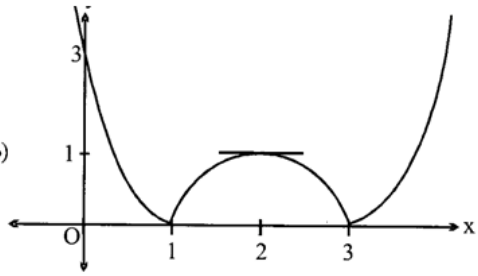
and decreasing ($\downarrow$) in $(-\infty, 1) \cup (2, 3)$

$g(x)$ is continuous $\forall\ x \in R$

But non-derivable at $x = 1, 3$

$g(x)$ has local maximum at $x = 2$,

and local minimum at $x = 1, 3$.



Graph of $g(x) = |x^2 - 4x + 3| = |(x-3)(x-1)|$

60. $2e^{x/y^2}\left(ydx - 2xdy\right) = -y^2 dy$

$2e^{x/y^2} \dfrac{\left(y^2 dx - 2xydy\right)}{y^4} = \dfrac{-y^3 dy}{y^4}$

$\int 2e^{x/y^2} d\left(x/y^2\right) = -\int \dfrac{dy}{y}$

$2e^{x/y^2} = -\ln y + c$

$2e^{x/y^2} + \ln y = c$

$y = 1,\ x = 0 \Rightarrow c = 2$

$2e^{x/y^2} + \ln y = 2$

$$y = e \Rightarrow x = -e^2 \ln(2)$$

61. $e^2 = 1 + \dfrac{b^2}{a^2}$ $\qquad (e')^2 = 1 + \dfrac{a^2}{b^2}$

$l = \dfrac{2b^2}{a}$ $\qquad l^1 = \dfrac{2a^2}{b}$

$\dfrac{a^2 + b^2}{a^2} = \dfrac{11}{14}.\dfrac{2b^2}{a}$ $\qquad 1 + \dfrac{a^2}{b^2} = \dfrac{11}{8}.\dfrac{2a^2}{b}$

$a^2 + b^2 = \dfrac{11}{7}ab^2$ $\qquad a^2 + b^2 = \dfrac{11}{4}a^2 b$

$\dfrac{11}{7}ab^2 = \dfrac{11}{4}a^2 b$ $\qquad a = \dfrac{4 \times 15}{7 \times 11} = \dfrac{65}{77}$

$4b = 7a$ $\qquad 77a + 44b = 130$

$\dfrac{16b^2}{49} + b^2 = \dfrac{11}{7}b^2 \dfrac{4b}{7}$

$\dfrac{16b^2 + 49b^2}{49} = \dfrac{44b^3}{49}$

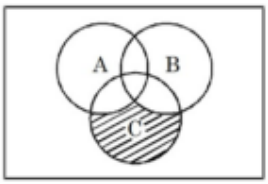
$b = \dfrac{65}{44}$

62. Let property, divisible by 5=A n(A) = 200

Divisible by 7 = B n(B) = 142

Divisible by 3 = C n(C) = 333

$n\left(\overline{A} \cap \overline{B} \cap C\right)$



$= 333 - \left[n(C \cap A) + n(C \cap B) - n(C \cap A \cap B)\right]$

$= 333 - [66 + 47 - 9] = 333 - 104$

$= 229$

$= \dfrac{229}{1000}$

63.

$$\frac{1}{abc}\begin{vmatrix} a^3+a & a^2b & a^2c \\ b^2a & b^3+b & b^2c \\ c^2a & c^2b & c^3+c \end{vmatrix} = \begin{vmatrix} a^2+1 & a^2 & a^2 \\ b^2 & b^2+1 & b^2 \\ c^2 & c^2 & c^2+1 \end{vmatrix}$$

$$= (1+a^2+b^2+c^2)\begin{vmatrix} 1 & 1 & 1 \\ b^2 & 1+b^2 & b^2 \\ c^2 & c^2 & 1+c^2 \end{vmatrix}$$

$$= (1+a^2+b^2+c^2) = k\sqrt{abc}$$

Now, $\dfrac{1+a^2+b^2+c^2}{4} \geq \sqrt{abc}$

$\Rightarrow$ $\dfrac{1+a^2+b^2+c^2}{\sqrt{abc}} = k \geq 4$

64. digit at unit place in

$11^{2012} + 23^{2012} - 3^{2012} = 1$

$$I = \int_0^1 \frac{1}{\sqrt{1-x^2} - x + \frac{1}{x}} =$$

$$= \int_0^1 \frac{x}{1-x^2+x\sqrt{1-x^2}}\,dx = \frac{\pi}{4} \text{ (Put } x = \sin\theta)$$

65. We have

$$t_n = \frac{1^2+2^2+3^2+.....+n^2}{1^3+2^3+3^3+.....+n^3} = \frac{\frac{1}{6}n(n+1)(2n+1)}{\frac{1}{4}n^2(n+1)^2}$$

$$= \frac{2(2n+1)}{3n(n+1)} = \frac{2}{3}\left[\frac{1}{n} + \frac{1}{n+1}\right]$$

$$\Rightarrow \sum_{k=1}^{n} t_k = \frac{2}{3}\left\{\left(1+\frac{1}{2}\right)+\left(\frac{1}{2}+\frac{1}{3}\right)+.....+\left(\frac{1}{n}+\frac{1}{n+1}\right)\right\}$$

$$= \frac{4}{3}H_n - \frac{2}{3}\frac{n}{n+1}$$

66. $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 3 & -4 \\ 8 & 4 & -\lambda \end{vmatrix} = 0 \Rightarrow 5\lambda - 72 = 0 \Rightarrow \lambda = \dfrac{72}{5}$

$$\Delta_3 = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 3 & 4 \\ 8 & 4 & 9+\mu \end{vmatrix} = 0 \Rightarrow -5\mu - 21 = 0 \Rightarrow \mu = \frac{-21}{5}$$

67. $\left[\cos^{-1} x\right] = 0 \; for \; \cos 1 < x \leq 1$

$\left[\cot^{-1} x\right] = 0 \; for \; \cot 1 < x < \infty$

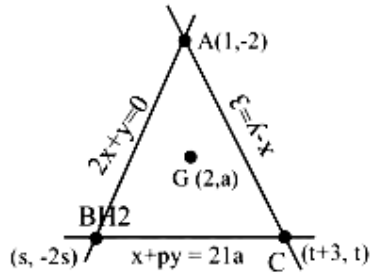
$\therefore \left[\cos^{-1} x\right] + \left[\cot^{-1} x\right] = 0 \; for \; x \in (\cot 1, 1]$

68. 2x+y=0 …….. (1)

x-y=0 …… (2)

x+py=21a …… (3)

solving (1) & (2) = A (1,2)



Centroid of triangle ABC is

$\left(\frac{4+s+t}{3},\frac{-2-2s+t}{3}\right)=(2,a) \qquad \Rightarrow s+t=2.....(4)$

$-2s+t=3a+2.....(5)$

Solving (4) 7 95) we get $\Rightarrow S=-a, t=2+a$

$B(-a,2a); C(a+5,a+2)$

$\because B, C \, lies \, on \;\; x+py=21a$

$\Rightarrow -a+2ap=21a \Rightarrow P=11$

$and \;\; pa+2p=20a-5$

$If \;\; p=11 \qquad \therefore 27-9a \Rightarrow a=3$

$\therefore B(-3,6), C(8,5), (BC)^2=(8+3)^2+(5-6)^2$

$\therefore Dis\tan ce \;\; (BC)^2=122$

69. We have $\frac{(x-1)^2}{\frac{25}{4}}+\frac{(y-2)^2}{\frac{25}{9}}=1$

So, min $\left((x-1)^2+(y-2)^2\right)=\frac{25}{9}$

and max $\left((x-1)^2+(y-2)^2\right)=\frac{25}{4}$

70. $x_1=a+2,\; x_2=b+2,\; x_3=c+2 \qquad n=3$

$\overline{x}=\frac{a+b+c}{3}+2$

$S.P=d=\sqrt{\frac{\sum(x_1-\overline{x})^2}{3}}$

$=\sqrt{\frac{\sum\left(a-\frac{a+b+c}{3}\right)^2}{3}}$

$27d^2=\sum(2a-b-c)^2$

$27d^2=6\sum a^2-6\sum ab$

$=6\left[a^2+b^2+c^2-ac\right]=2a^2+2c^2-2ac$

$9d^2=\left[a^2+c^2-ac\right]=2a^2+2c^2-2ac$

$b^2=a^2+c^2+2ac$

$b^2+9d^2=3\left(a^2+c^2\right)$

71.

$$\lim_{x\to 0}\frac{2\sin^2\left(\frac{ax^2+bx+c}{2}\right)}{\left(\frac{(ax^2+bx+c)}{2}\right)^2}\frac{(ax^2+bx+c)^2}{4(x-\alpha)^2};\quad \lim_{x\to\alpha}\frac{2a^2\left(x^2+\frac{b}{a}x+\frac{c}{a}\right)\left(x^2+\frac{b}{a}x+\frac{c}{a}\right)}{4(x-\alpha)\qquad(x-\alpha)}$$

$$\lim_{x\to\alpha}\frac{a^2(x-\alpha)^2\ \ (x-\beta)^2}{2(x-\alpha)^2};\quad \lim_{x\to\alpha}\frac{a^2}{2}(\alpha-\beta)^2\Rightarrow C$$

72. $f(x)+\int_0^x f(t)\sqrt{1-\left(\log_e f(t)\right)^2}\,dt=e$

By Leibnitz condition

$f'(x)+f(x)\sqrt{1-\left(\log_e f(x)\right)^2}(1)-0=0$ $\qquad$ $f'(x)+f(x)\sqrt{1-\left(\log_e f(x)\right)^2}=0$

$f'(x)=-f(x)\sqrt{1-\left(\log_e f(x)\right)^2}$ $\qquad$ $\frac{1}{\sqrt{1-\left(\log_e f(x)\right)^2}}\frac{f^1(x)}{f(x)}=-1$

Integrate on both sides

$\int\frac{1}{\sqrt{1-\left(\log_e f(x)\right)^2}}\frac{f^1(x)}{f(x)}dx=-\int dx$ $\qquad$ $\log_e f(x)=t$

Differentiate on both sides w.r.t to x $\qquad$ $\frac{f'(x)}{f(x)}dx=dt$ $\qquad$ $\int\frac{1}{\sqrt{1-t^2}}dt-\int dx$

$\sin^{-1}(t)=-x+c$ $\qquad$ $\sin^{-1}\left(\log_e f(x)\right)=-x+c$.....(1)

Put $x=0\Rightarrow\sin^{-1}\left(\log_e f(x)\right)=0+c$ $\qquad$ $\because f(0)=e$ $\qquad$ $\sin^{-1}(1)=c\Rightarrow c=\frac{\pi}{2}$

From (1)

$\sin^{-1}\left(\log_e f(x)\right)=-x+\frac{\pi}{2}\log_e f(x)=\sin\left(\frac{\pi}{2}-x\right)$ $\qquad$ $\log_e f(x)=\cos x$

Put $x=\frac{\pi}{2}$ $\qquad$ $\log_e f\left(\frac{\pi}{2}\right)=\cos\frac{\pi}{6}=\frac{\sqrt{3}}{2}$

$6\log_e f\left(\frac{\pi}{2}\right)=6\left(\frac{\sqrt{3}}{2}\right)=3\sqrt{3}$

S.O.B $\qquad$ $\left(6\log_e f\left(\frac{\pi}{2}\right)\right)^2=\left(3\sqrt{3}\right)^2=27$

74.

$|\vec{c}|^2=4(\vec{a}\times\vec{b})^2+9b^2=4(a^2b^2-(\vec{a}\cdot\vec{b})^2)+9b^2=192$

$\vec{c}+3\vec{b}=2\vec{a}\times\vec{b}\Rightarrow c^2+9b^2+6\vec{b}\cdot\vec{c}=4(a^2b^2-(\vec{a}\cdot\vec{b})^2)$

$\Rightarrow\quad 6\cdot 4\cdot\sqrt{192}\cos\theta=-288\quad\Rightarrow\cos\theta=\frac{-\sqrt{3}}{2}$

75.

$$y = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$$

$$\frac{dy}{dx} = 5(2x - 3\pi)^4 + \frac{4}{3} - \sin x$$

$$\therefore \quad g'(y) = \frac{dx}{dy} = \frac{1}{5(2x - 3\pi)^4 + \frac{4}{3} - \sin x}$$

When $y = 2\pi$

$$2\pi = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$$

$x = \frac{3\pi}{2}$ is the only solution

$$g'(2\pi) = \frac{1}{1 + \frac{4}{3}} = \frac{3}{7}$$